Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Zerwekh, Joseph E Ph.D.		
Project Title:	Prevention of Microgravity-Induced Stone Risk by KMgCitrate		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-8885	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Antich, Peter (The University of Texas Southwestern Medical Center at Dallas) Pak, Charles (The University of Texas Southwestern Medical Center at Dallas) Odvina, Clarita (The University of Texas Southwestern Medical Center at Dallas)		
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Task Description:	The hypothesis to be tested in this project is that potassium magnesium citrate supplementation will attenuate the increased risk for stone formation and diminish microgravity-induced bone loss. This hypothesis will be tested during five weeks of bed rest in normal volunteers through three specific aims: Specific aim 1: Assess the efficacy of supplementation with potassium magnesium citrate (KMgCit) in preventing microgravity-induced increased risk of stone formation. The hypothesis to be tested is that supplementation with KMgCit will avert the increase in urinary stone forming risk during bed rest by increasing urinary pH, promoting magnesium complexation of oxalate, and raising urinary citrate, an established inhibitor of crystal agglomeration. Specific aim 2: Evaluate the effect of KMgCit supplementation in averting diminished muscle magnesium and potassium concentrations that may occur during microgravity-induced muscle atrophy. The hypothesis to be tested is that by providing adequate potassium and magnesium supplementation, the loss of these elements from muscle during prolonged bed rest will be averted. Specific aim 3: Assess the efficacy of supplementation with KMgCit in reducing microgravity-induced increases in bone resorption and urinary calcium. The hypothesis to be tested is that during five weeks of bed rest where bone loss is known to occur, supplementation with KMgCit will prevent the negative impact of low alkali intake and mild acidosis on bone resorption, urinary calcium loss, and further reduce the increased risk for calcium oxalate stone formation. The study was completed in 20 normal subjects, 9 of whom received placebo and 11 received active drug. Compared to placebo, KMgCit administration resulted in significant increases in urinary pH, potassium, magnesium, and citrate while urinary total titratable acidity and ammonium significantly decreased. These changes in urinary composition resulted in a significant decline in the relative saturation of urine with respect to calcium oxalate and undissociated uric acid and thus reduced the propensity for stone formation with these two salts. There were no significant changes in brushite saturation. In contrast, the saturation of calcium oxalate and uric acid increased in the placebo-treated subjects. Thus, KMgCit decreased the risk for renal stone formation despite comparable increases in urinary calcium during bed rest in both groups. Changes in muscle K and Mg were less informative. There was no significant change in muscle K or Mg in the placebo group but muscle Mg significantly declined in the KMgCit-treated subjects, a response that is opposite to what was anticipated. Therefore, it appears that KMgCit does not have a pronounced effect at muscle. Lastly, there were no significant differences in the bone resorption markers deoxypyridinoline, n-telopeptide, or serum carboxytelopeptide indicating that KMgCit was ineffective in preventing increased bone resorption resulting from skeletal immobilization. Thus, our primary hypothesis that provision of alkali as KMgCit would serve to reduce the increased risk of renal stone formation encountered during spaceflight, has been supported. These findings also suggest that this countermeasure might also be useful in preventing renal stone disease in Earth-based renal stone formers. As this study is now completed, there is no proposed research plan for the coming year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Nephrolithiasis is a significant medical condition that was responsible for 1.3 million physician office visits in 1997 and an estimated 290,000 hospital discharges in 1999 due to a stone-related problem. It is apparent that any advances made in our understanding of the metabolic, nutritional, and environmental factors that contribute to increased stone risk in space will have direct application to this pervasive problem on Earth. Although there may be differences in the etiology of hypercalciuria between astronauts and Earth-based stone formers, there are striking similarities in the changes that can occur in the urinary environment predisposing to stones. Nutritional, as well as pharmacologic based countermeasures, that may prove to be effective in reducing stone risk in astronauts, would be a welcomed additional tool for the management of Earth-based disease. The results from this study support the use of KMgCitrate on Earth for those individuals who suffer from unduly acidic urine and/or uric acid stone formation. This drug should prove to be useful in reducing the prevalence and in preventing the recurrence of nephrolithiasis. Unfortunately, our study did not demonstrate a beneficial effect of our countermeasure in reducing bone mineral loss as assessed by bone biochemical markers. Although other studies of the benefits of alkali therapy in postmenopausal women have demonstrated a beneficial effect on calcium economy and reduced bone loss, under the rigors of immobilization, alkali therapy may not be an effective countermeasure. Thus, KMgCitrate could still be useful for the 25 million Americans who are osteopenic or osteoporotic. Additional studies will be needed to address this issue.
Task Progress:	We have completed our proposed bed rest study. Due to restrictions in bed utilization by the GCRC from 2 beds for this study to one bed, we had to extend our timeline by 1 year in order to meet our proposed number of study subjects of 20. This has been accomplished.
Bibliography Type:	Description: (Last Updated: 02/05/2007)
Abstracts for Journals and Proceedings	Zerwekh JE, Odvina CV, Pak CYC. "Prevention of microgravity-induced renal stone disease by potassium-magnesium-citrate (KMgCit)." Berlin Heidelberg, Germany, 2004 May. Urological Research. 2004 May;32:164. , May-2004
Articles in Peer-reviewed Journals	Kondrashov V, Rothenberg SJ, Chettle D, Zerwekh J. "Evaluation of potentially significant increase of lead in the blood during long-term bed rest and space flight." Physiol Meas. 2005 Feb;26(1):1-12. PMID: 15742874 , Feb-2005
Articles in Peer-reviewed Journals	Zerwekh, J E. "Nutrition and renal stone disease in space." Nutrition. 2002 October;18:857-863. , Oct-2002
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